Italian Named Entity Recognition with Transformer Models

UmBERTo vs XLM-RoBERTa on NERMuD 2023

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The question

Does an Italian-specific model beat a multilingual one for Italian NER?

What is NER?

The operation for **Pope Leo** will take place at the **Agostino Gemelli Hospital** in **Rome**.

PER Person

ORG Organization

LOC Location

The two models

Both are RoBERTa-based Transformers: self-attention looks at all the words at once

UmBERTo
Italian-specific

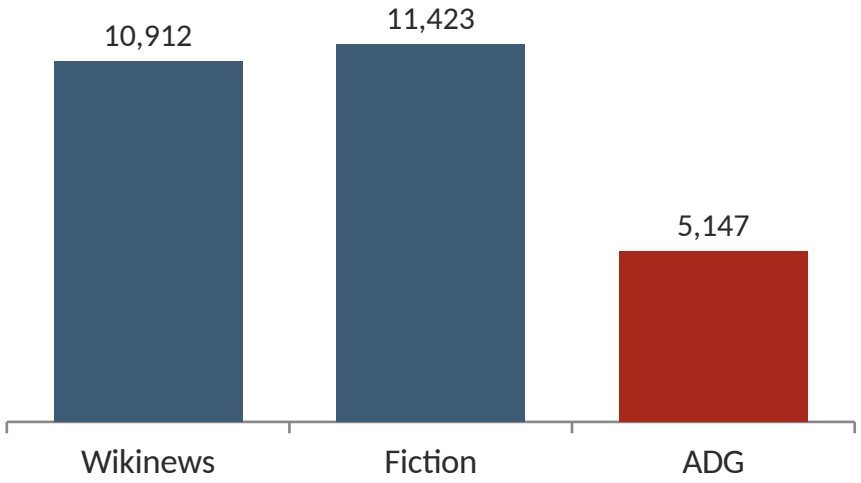
32k	vocabulary
Italian	OSCAR corpus
WWM	whole word masking

XLM-RoBERTa
Multilingual

250k	vocabulary
100	languages, 2.5 TB
MLM	masked LM only

Dataset: NERMuD 2023

Training examples per domain



News · Literature · De Gasperi speeches

BIO labels

Pope	Leo	visited	Rome
B-PER	I-PER	O	B-LOC

- B = beginning of an entity
- I = inside an entity
- O = outside any entity

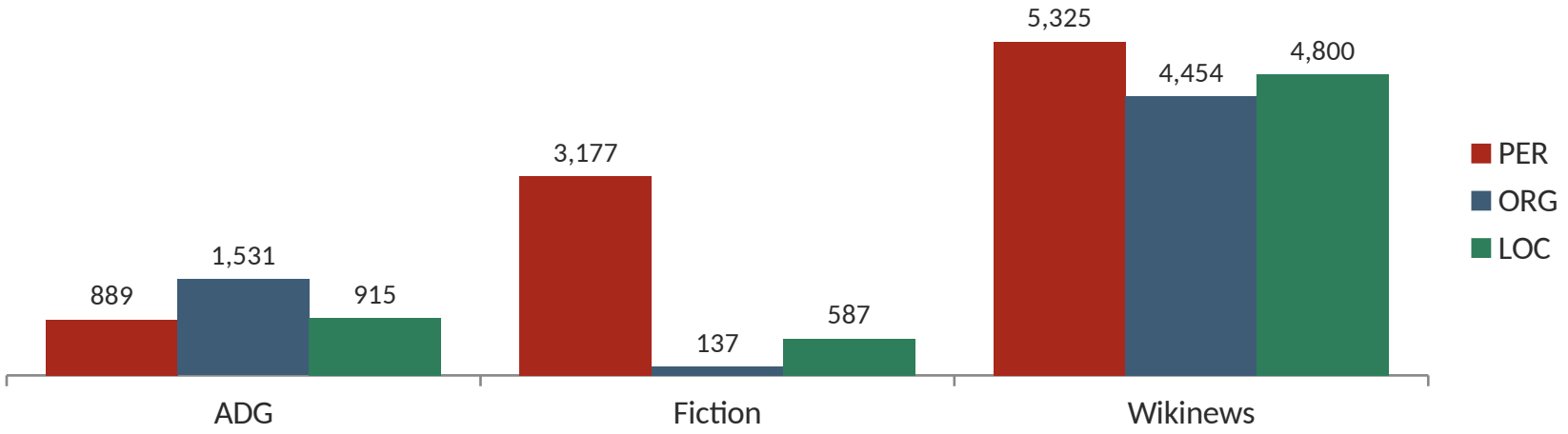
27,482 train · 4,767 dev · 4,126 test

Entities in the training set

9,391 PER

6,122 ORG

6,302 LOC



Same training protocol

2e-5	8	3	256	42
learning rate	effective batch size	epochs	max length	random seed

- Best checkpoint chosen on the dev set
- Evaluated with seqeval: entity-level precision, recall, F1
- An entity is correct only if boundaries and type are both right
- F1 as main metric: most tokens are "O", so accuracy misleads

Global result: UmBERTo wins

0.8945

UmBERTo · test F1

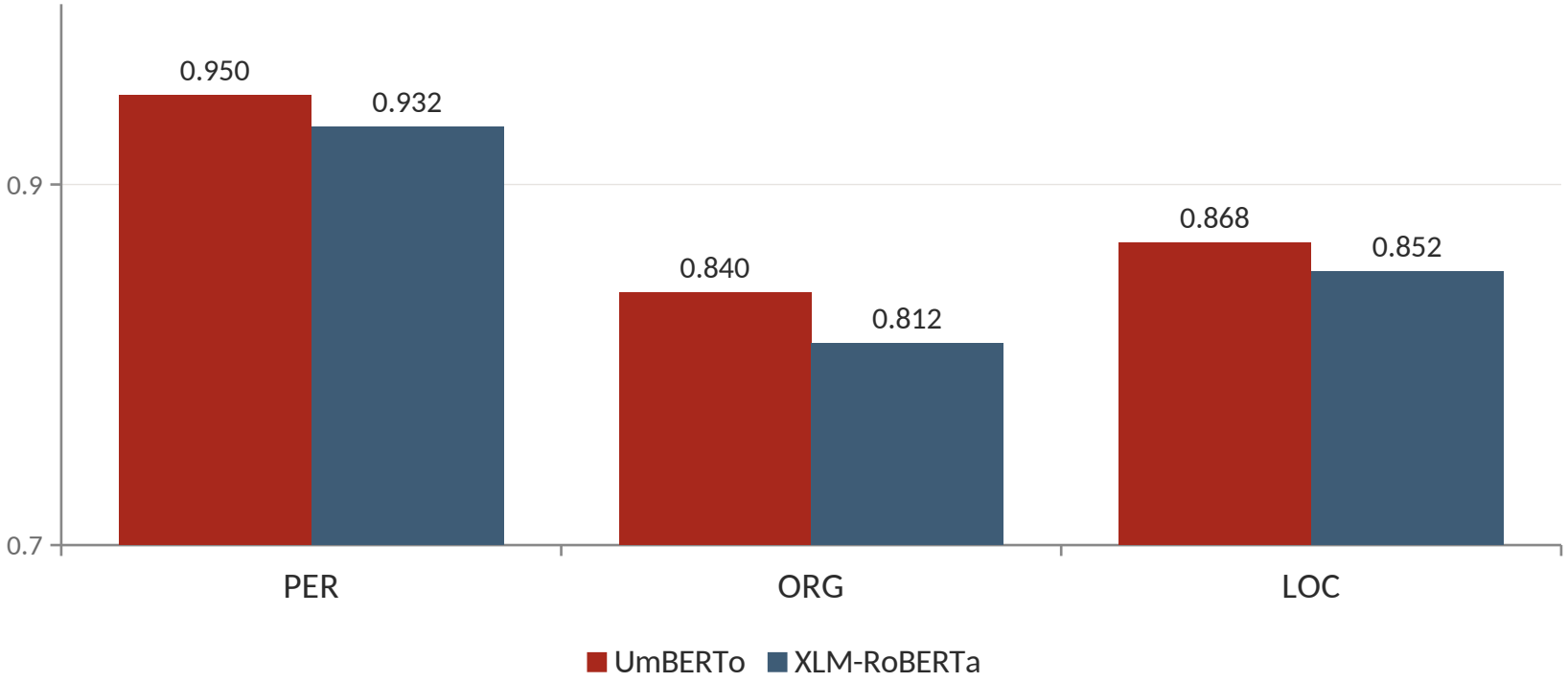
0.8743

XLM-RoBERTa · test F1

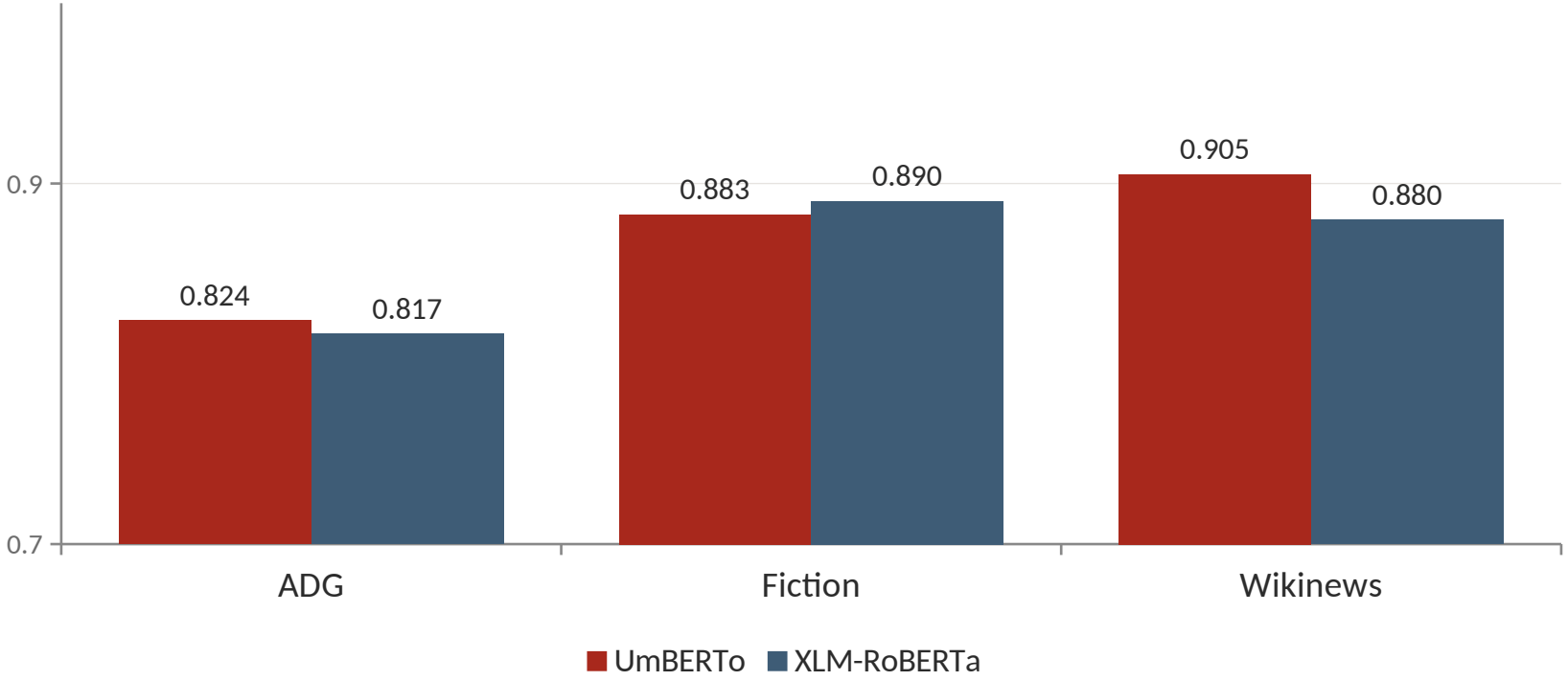
Gap of ≈ 2 F1 points, consistent in precision and recall

For reference: best NERMuD 2023 systems score around 0.90 F1

By entity type



By domain


0.9
0.7
0.824
0.817
0.883
0.890
0.905
0.880
ADG
Fiction
Wikinews
UmBERTo
XLM-RoBERTa

Takeaways

The answer: yes, but...

- UmBERTo is better overall
- XLM-R is better on fiction
- Gap is not uniform across domains

Limitations & next steps

- Single run, no significance test
- Models differ beyond language
- Next: multiple seeds, larger models, other datasets

Course coverage

Covered

- Downstream tasks: fine-tuning Transformers
- Transfer learning
- Evaluation of NLP applications
- Corpora and annotations (BIO)
- Introduction to deep learning

Only touched on

- Transformer architecture
- Tokenisation (subwords only)
- Embeddings (contextual, indirectly)
- Sequence labelling, as in POS tagging

Not covered

- N-gram language models
- Speech and phonetics
- Morphology
- Syntax: grammars, parsing
- Semantics: WordNet, WSD
- Generative models, prompting, agents